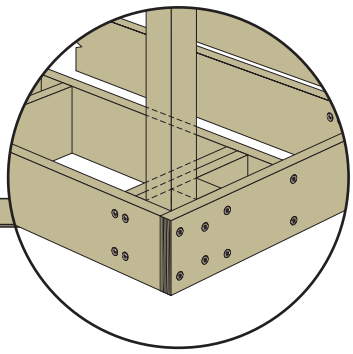
Guard Post to Deck Framing

The guard post connection is a crucial connection pertaining to deck safety and is often inadequately constructed. In order to provide the required load resistance at the guard, the post must not only be fastened to the rim joist, but also tied back into the joist framing. Machine bolts through the post and rim joist alone do not typically meet the performance requirements of the code. The details on pp. 39–42 have been shown through testing or analysis to resist the required 200 lb. concentrated load (or load of 50 lb. per linear foot per IBC when guardrail spacings do not exceed 48") applied in an outward direction for maximum guard heights of 36" and 42" above the deck surface. Guard post to deck attachments comply with IRC section R507.10.1.1, which prohibits connections only relying on end grain withdrawal.

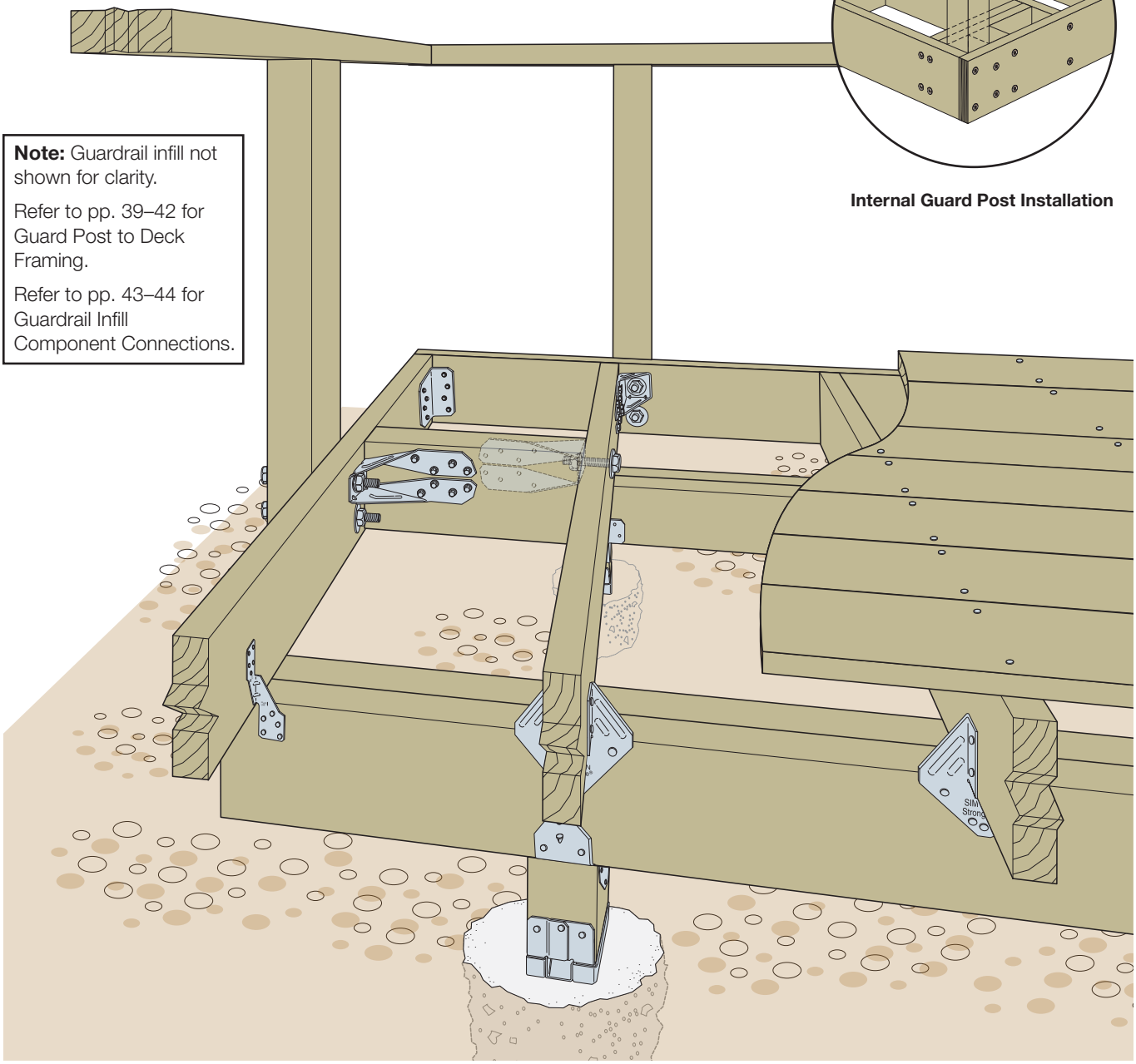


Internal Guard Post Installation

Note: Guardrail infill not shown for clarity.

Refer to pp. 39–42 for Guard Post to Deck Framing.

Refer to pp. 43–44 for Guardrail Infill Component Connections.

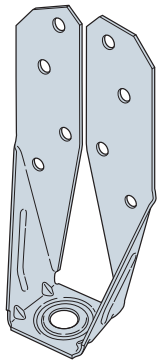


For more information on this application, see Simpson Strong-Tie technical bulletin T-C-GRDRLPST.

Guard Post to Deck Framing — Holdown Details

Simpson Strong-Tie® Solutions

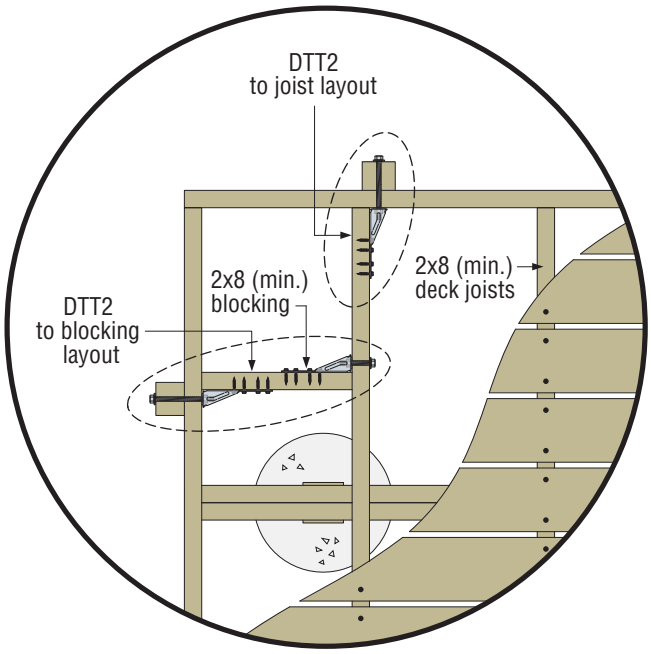
For 36"-Tall Guard Posts



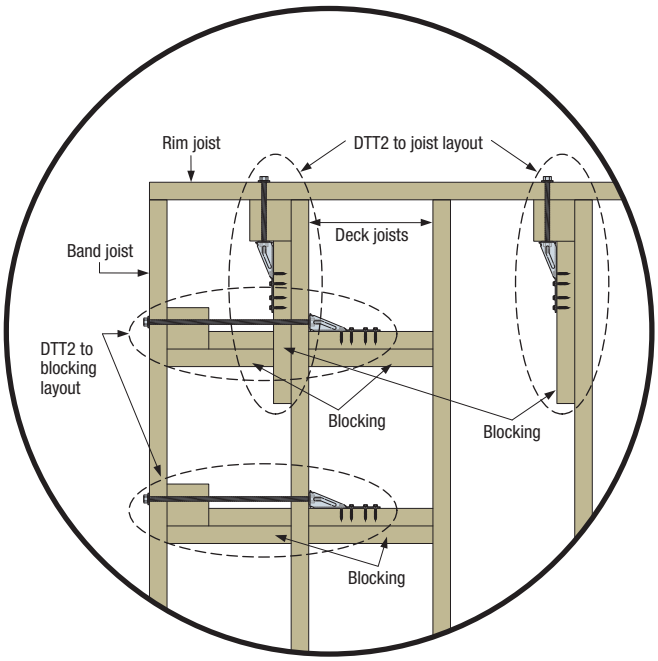
DTT2 Deck Tension Tie
(available with ZMAX® coating or in stainless steel, incudes SDS Heavy-Duty Connector screws)
US Patent 8,555,580B2



SDS Heavy-Duty Connector Screw



DTT2 Attachment of Guardrail
Post *Outside* Rim/Joist



DTT2 Attachment of Guardrail
Post *Inside* Rim/Joist

Note: Guard post to deck attachments comply with IRC section R507.10.1.1, which prohibits connections only relying on end grain withdrawal.

Building Code Requirements

When required

- Guards shall be located along open-sided walking surfaces, including floors, stairs, ramps and landings, that are located more than 30" measured vertically to the floor or grade below at any point within 36" horizontally to the edge of the open side.

IRC 2018/2021/2024 Section R312.1.1
IRC 2024 Section R321.1.1
IBC 2018/2021/2024 Section 1015.2

Height

- Guards shall be a minimum of 36" tall (IRC) or up to 42" tall for certain occupancies (IBC).

IRC 2018/2021/2024 Section R312.1.2
IRC 2024 Section R321.1.2
IBC 2018/2021/2024 Section 1015.3

Load Resistance

- Guards and handrails shall be able to resist a single concentrated load of 200 lb., applied in any direction at any point along the top. (IRC 2021/2024 simplifies the load direction to downward and horizontally outward for the top of a guard not required to serve as a handrail).

IRC 2018/2021/2024 Table R301.5
IBC 2018 Section 1607.8.1.1
IBC 2021/2024 Section 1607.9

- Handrails and guards shall be designed to resist a load of 50 lb. per linear foot.

IBC 2018 Section 1607.8.1
IBC 2021/2024 Section 1607.9
(not required for one- and two-family dwellings)

- Guardrail infill components shall withstand a horizontal load of 50 lb. over an area of one square foot.

IRC 2018/2021/2024 Table R301.5
IBC 2018 Section 1607.8.1.2
IBC 2021/2024 Section 1607.9.1.2

Opening Limitations

- Required guardrail infill shall not have an opening that allows passage of a sphere 4" in diameter.

IBC 2018/2021 Section R321.1.3
IRC 2024 Section R321.1.3
IBC 2018/2021/2024 Section 1015.4

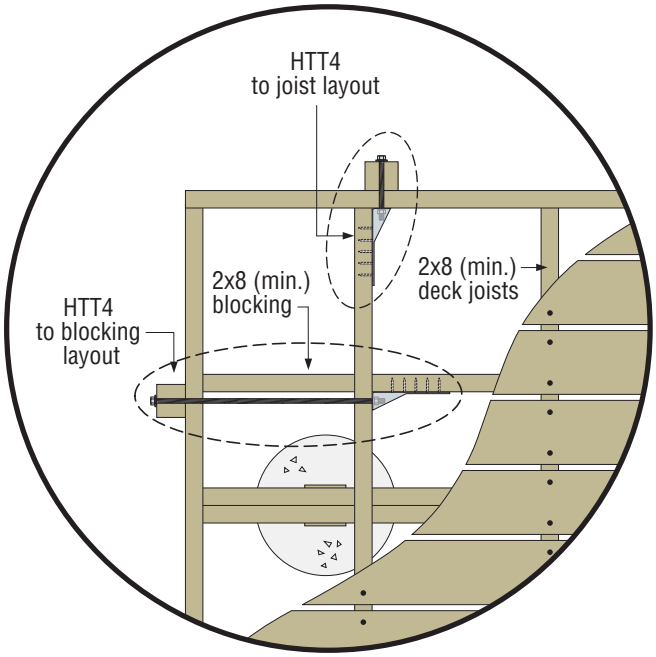
For more information on connecting posts inside the rim joist using the DTT2 or HTT4, see Simpson Strong-Tie technical bulletin T-C-GRDRLPST.

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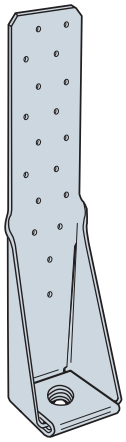
Guard Post to Deck Framing — Holdown Details

For 42"-Tall (maximum) Guard Posts

Note: The DTT2 may also be used to secure 42" guard posts with 2x10 (min.) deck joists.



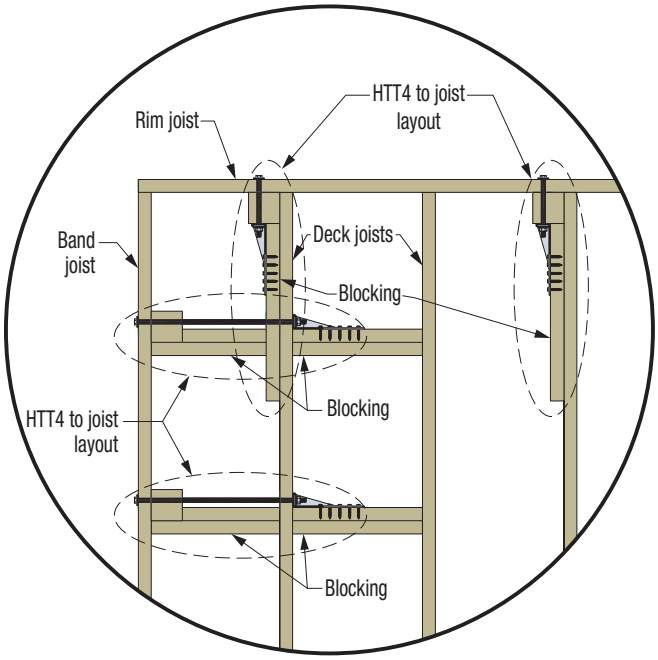
HTT4 Attachment of Guardrail
Post *Outside* Rim/Joist



HTT4/HTT4HDG Tension Tie
Available with hot-dip galvanized coating.



Strong-Drive® SD Connector Screw



HTT4 Attachment of Guardrail
Post *Inside* Rim/Joist
(not all fastening for joists and blocking shown for clarity)



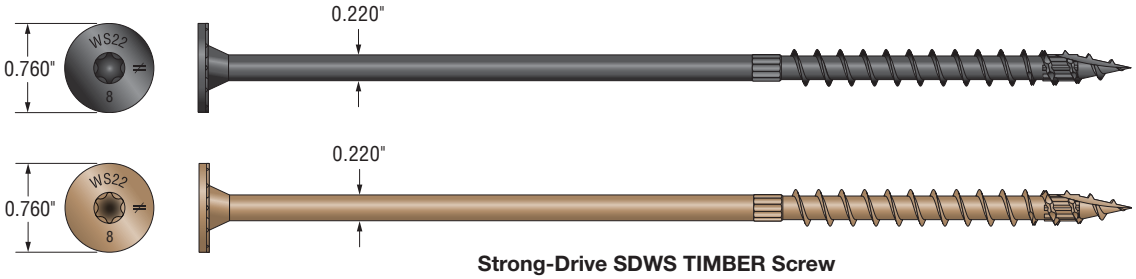
ATR All Threaded Rod/
RFB Retrofit Bolt
Available with hot-dip galvanized coating or in Type 316 stainless steel.
(an additional nut will be required for the HTT4 application)

For more information on connecting posts inside the rim joist using the DTT2 or HTT4, see Simpson Strong-Tie technical bulletin T-C-GRDRLPST.

Guard Post to Deck Framing — Fastener Details

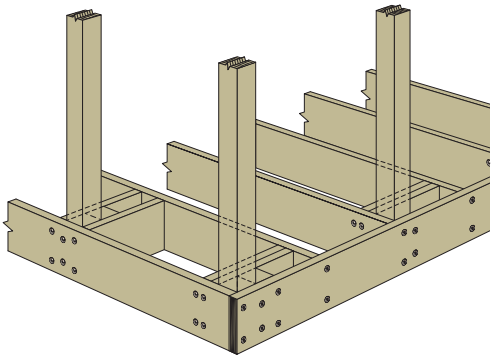
For 36"- and 42"-Tall Guard Posts

As an alternative code-compliant solution for attaching wood guard posts without the use of typical hardware and through-bolts, the guard posts and the framing elements can be fastened with Strong-Drive® SDWS™ Timber screws. The SDWS Timber screws are code report listed in IAPMO UES ER-192. The solutions below have been tested and verified for 36"-tall guard posts constructed with DFL, HF and SP framing. These solutions have also been tested and verified for 42"-tall guard posts constructed with DFL and SP framing only. For more information, please refer to *Fastening Systems Technical Guide* or T-F-GRDPSTRL.

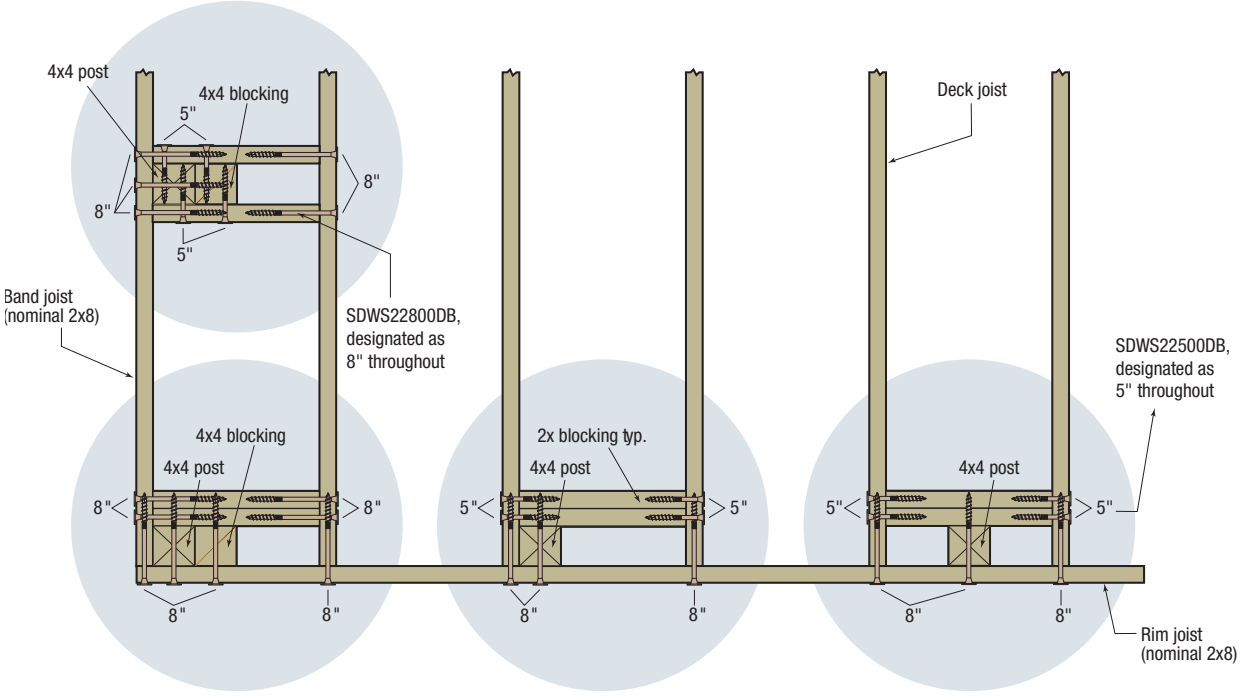


Strong-Drive SDWS TIMBER Screw
(SDWS22xxxDB: Tan Double-Barrier Coating; SDWS22xxxDBB: Black Double-Barrier Coating)

Note: For this installation, only SDWS22500DB/DBB and SDWS22800DB/DBB fasteners are to be used and installed per Simpson Strong-Tie details. Test results have shown that generic fasteners do not provide sufficient withdrawal resistance, pull-through resistance or shear strength.



Internal Guard Post Installations Using Strong-Drive SDWS Timber Screws



Plan View Showing Details of Four Connections Using Strong-Drive SDWS Timber Screws
(SDWS22xxxDB Tan shown, SDWS22xxxDBB Black similar; other connection hardware not shown for clarity)

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